

Exercise B

Let us consider the inequality $B \preceq A$, between two squared matrices A and B of dim $p \times p$. The symbol $\preceq$ means that $A - B$ is positive semi-definite.

Assume (x_i, y_i) is linear, therefore $y = X\beta + \varepsilon$, with $\varepsilon_i \stackrel{iid}{\sim} \mathcal{N}$, with $E[\varepsilon_i] = 0$, and $\text{var}(\varepsilon_i) = \sigma^2$.

Let $\hat{\beta}_{OLS} = (X^T X)^{-1} X^T y$ and let $\tilde{\beta} = Ay$ such that $\tilde{\beta} \neq \hat{\beta}_{OLS}$ and $E[\tilde{\beta}] = \beta$. Then

Decompose

$$\text{Let } C = A - (X^T X)^{-1} X^T \Rightarrow A = (X^T X)^{-1} X^T + C.$$

By assumption, we have that $E[\tilde{\beta}] = \beta$ and

$$E[\tilde{\beta}] = E[Ay] = AX\beta = \cancel{(X^T X)^{-1} X^T} X\beta + C X\beta = \beta(I + CX).$$

Hence, we must have that $CX = 0$. Now, note that

$$\tilde{V} = \text{var}(Ay) = \sigma^2 AA^T =$$

$$= \sigma^2 [(X^T X)^{-1} X^T + C] [(X^T X)^{-1} X^T + C]^T$$

$$= \sigma^2 [(X^T X)^{-1} X^T + C] [X(X^T X)^{-1} + C^T]$$

$$= \sigma^2 [\cancel{(X^T X)^{-1} X^T} X (X^T X)^{-1} + \underbrace{(X^T X)^{-1} X^T C^T}_{(CX)^T = 0} + \underbrace{CX (X^T X)^{-1}}_0 + CC^T]$$

$$= \sigma^2 (X^T X)^{-1} + \sigma^2 CC^T$$

$$= V + \sigma^2 CC^T$$

$\Rightarrow \tilde{V} - V = \sigma^2 CC^T$, which is positive semi-definite (same proof of the matrix $(X^T X)$).